

Requirements Specification

Team Project Module · National College of Ireland · Individual Project

Project Name: PrepCast

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Module / Cohort: Team Project Module (Individual)

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Functional Requirements

ID	User Story	Category	Priority	Acceptance Criteria
FR-001	As a manager, I want the system to use ML model prediction, so that the forecasts that I receive are accurate and based on historical sales.	Core	MUST	Given that several possible regression models are available, When evaluated on a time-based train/test split, Then the model with the best evaluation metrics is chosen as the core brain of the system.
FR-002	As a manager, I want the system to take promotions and seasonality into account, so that the forecasts always give an accurate output during busy weeks.	Core	MUST	Given that some weeks have promotions, When the forecast is created, Then a week with a promotion gives a higher predicted demand than a non-promotional week.
FR-003	As a manager, I want to select my location centre, so that I can only see my location's forecasts.	UI/UX	MUST	Given that the app is open, When the manager selects their centre location, Then only that location's forecasts for the next week are shown.
FR-004	As a manager, I want to see next week's forecasted demand and recommended prep per meal, so that I can plan for the expected demand.	Core	MUST	Given that the manager selected their centre, When they view the dashboard, Then they see the per-meal forecast and recommended prep (forecast + safety buffer) for the next week displayed.
FR-005	As a manager, I want to save and adjust my prep plan, so that I can factor in local knowledge.	Data	SHOULD	Given a recommended plan, When the manager edits a meal quantity and save, then the plan is saved to the database.
FR-006	As a manager, I want to record actual meal orders, so that the model can learn from current demand.	Data	SHOULD	Given that the week has ended, When the manager records the actual orders, Then they are saved to the database and available for future forecasting.
FR-007	As a manager, I want the forecasts to update after new data is uploaded, so that the forecasting is current and up-to-date.	Core	COULD	Given that the latest sales are saved, When a new forecast is generated, Then it uses the updated data.
FR-008	As a manager, I would like to see the forecast for next week compared to 'last week' sales, so that I can trust the system.	UI/UX	COULD	Given a forecast plus the 'last week' sales, When the data is displayed, Then the manager can see a comparison view.

Non-Functional Requirements

ID	Category	Requirement	Priority	Measurable Target
NFR-001	Performance	Recommendations returned quickly	MUST	Returns in under 2 seconds
NFR-002	Usability	Manager accesses prep plan in few clicks	SHOULD	3 clicks or fewer
NFR-003	Compatibility	Runs on a standard laptop browser	SHOULD	Runs on Chrome, Edge and Safari
NFR-004	Performance	Model accuracy	SHOULD	Model achieves $R^2 \geq 0.80$